

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 13 · LECTURE NOTES

Claiming a Mechanism Works

The solution concept, the agent model, and the impossibility you chose.

MODULE OUTCOMES

State what a mechanism claim must specify.

Identify the standard overclaims.

Explain why a manipulation claim needs an exhibit.

Place this course in the semester and the program.

Evaluate a deployed mechanism.

1 What a claim must name

A MECHANISM CLAIM NAMES A SOLUTION CONCEPT, A MODEL OF THE AGENTS, AND THE IMPOSSIBILITY IT LIVES WITH

The solution concept — dominant strategy, Bayes-Nash, or ex-post equilibrium — because they assume wildly different things about what each agent knows about the others (Module 02 §2), and a Bayes-Nash guarantee requires a common prior that may not exist.

The agent model — that agents know their own valuations, can compute best responses, do not collude, and have no interests outside the mechanism — four assumptions, each of which fails somewhere (Module 07 §4's measured shading).

And the impossibility it lives with — every mechanism gives something up (Module 06 §4's Myerson-Satterthwaite, Module 10 §3's Gibbard-Satterthwaite) — and naming which one is what makes a design defensible rather than merely described.

Plus the benchmark and the approximation ratio, wherever the allocation is computed approximately (Module 08) — because a ratio stated without the optimum it is a ratio to is not a number, which is CSCE 629's discipline arriving here.

The claim	The correction
'The mechanism is incentive compatible.'	In which solution concept, and under what model of the agents? (§1.)
'It is strategy-proof.'	For which side? (Module 09 §2 — deferred acceptance is strategy-proof for the proposers only.)
'It is optimal.'	For which objective — welfare, revenue, or fairness? See the note.

The claim	The correction
'The allocation is fair.'	Which fairness notion? (Module 11 §1's five, which can conflict.)
'Agents will play the equilibrium.'	Which equilibrium, and can they compute it? (Module 03, and Module 12's answer.)
'The mechanism is manipulable.'	Show the manipulation. (§2.)

'It is optimal' is the one that hides the most — efficiency, revenue, and fairness are different objectives with genuinely different optimal mechanisms, and a revenue-optimal auction is deliberately inefficient (Module 07 §3). Each correction above asks for a qualifier the claim omitted, which is the same pattern as the other two Semester 12 courses' closing tables.

2 Manipulations need exhibits

'IT IS MANIPULABLE' WITHOUT A SPECIFIC PROFITABLE MISREPORT IS NOT A FINDING

Gibbard-Satterthwaite already tells you that almost every mechanism without money is manipulable (Module 10 §3) — so the bare general claim carries no information at all and is not worth making.

What is informative is: which agent, with what true preferences, submitting what misreport, gaining how much — four things, and together they are an exhibit somebody else can check and reproduce.

And it is what distinguishes a serious criticism from a gesture: a mechanism manipulable only by an agent with full knowledge of everybody else's reports is practically very different from one manipulable by a simple rule any participant could follow — and only the exhibit tells you which you have.

Which is why Project 2 requires either a proof of truthfulness or a concrete manipulation — and it is worth noticing that the proof is frequently easier to produce than a convincing exhibit, which is a good sign about the standard of evidence being asked for.

3 The semester and the program

Course	Who sets the rules	The adaptation required
CSCE 640	Physics.	Design within a gate set, a measurement rule, and a noise model you cannot change — and state the resource cost.
CSCE 628	Biology.	Validate by controls and proxies, because no ground truth is available to check against.
CSCE 717 (this one)	Strategic agents.	Make honesty optimal, because you cannot observe the true input and never will.

In all three courses the true state is unobservable and the model is not yours to choose — the amplitudes cannot be read, the ground truth is not available, the valuations are private — **which is the semester's common structure and its transferable lesson. Unobservable truth is what the three courses actually share**, and each answers it differently: interference engineering, negative controls, and incentive compatibility are three responses to the same predicament.

4 Evaluating a deployed mechanism

- **What is the designer's objective? — and is it the one you assumed? A revenue-maximising auction is deliberately inefficient** (Module 07 §3), and reading it as a failed attempt at efficiency is a misdiagnosis.
- **What is reported, and what is the incentive to misreport it?** (Module 01 §2's table) — **which is the first thing to ask of any system** and takes thirty seconds.
- **Which solution concept does the guarantee use, and can the agents actually reach it?** (Module 03's hardness, and Module 12's answer about what they reach instead).
- **Which impossibility is it living with, and is that the right thing to give up in this setting?** (Module 06 §4, Module 10 §3) — since something must be given up, the question is only whether the choice suits.
- **And what happens under collusion, false identities, and repeated play? — three things the single-shot model omits and that practice does not. They are where deployed mechanisms actually fail**, rather than at the points the theory worries about.

WHERE THIS COURSE LEAVES YOU

You can recognise a strategic input when you see one (Module 01) — **which is the single most transferable thing in this course, and applies to systems you build that have nothing to do with auctions:** priority fields, quotas, resource requests, and every metric that determines an allocation.

You can measure what decentralisation costs (Module 04's price of anarchy) **and decide whether to intervene** — **which is a decision engineers make constantly by instinct** and can now make with a number.

And you can design a mechanism, state its guarantee precisely, and name what it gives up (Modules 06 through 11) — **which is the engineering skill this course exists for**, and is what Project 2 assesses.

The closing rule is the program's, unchanged across thirty-six courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it is the solution concept, the agent model, and the impossibility — because 'incentive compatible' is a claim about a model of people, and the people are the part the model is least sure about.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Roughgarden — Twenty Lectures, the closing lectures (free)	§§1 and 4 — what the guarantees mean and where they stop, stated by somebody careful about it.
Milgrom — Discovering Prices	§4 — a deployed mechanism described by its designer, including what had to be given up and why. Library copy.
Rothkopf — Thirteen reasons why the Vickrey-Clarke-Groves process is not practical	§§1 and 4 — the practical objections to the theoretically optimal mechanism, enumerated.
The ACM EC proceedings (free preprints)	§§1 and 2 — current work, and the better papers state all three of §1's items in the abstract.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the three things a mechanism claim must name**, for a mechanism you have seen.
- 2 **Find three published mechanism claims** and check which of the three each names.
- 3 **Correct six overclaims** in your own words.

- 4 **Take a 'strategy-proof' claim** and identify the side.
- 5 **Produce a concrete manipulation exhibit** for one mechanism, with all four components.
- 6 **Compare its difficulty** to proving truthfulness for a different one.
- 7 **State each Semester 12 course's unobservable truth.**
- 8 **Evaluate one deployed mechanism** against §4's five questions.
- 9 **Find a mechanism that failed in practice** for one of §4's last three reasons.
- 10 **Project 2 is now due.** Submit the mechanism, its incentive proof or concrete manipulation, the approximation ratio against a named benchmark, the strategic simulation compared to the honest baseline, and the impossibility you chose to live with.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three things a mechanism claim must specify.
- 2 What fourth thing is needed when the allocation is approximate?
- 3 Correct six standard overclaims.
- 4 Which overclaim hides the most, and why?
- 5 Why is a general manipulability claim uninformative?
- 6 What four things make a manipulation exhibit?
- 7 Why does the exhibit matter practically?
- 8 State each Semester 12 course's external model and unobservable truth.
- 9 Give the five questions for evaluating a deployed mechanism.
- 10 Where do deployed mechanisms actually fail?